STAEformer 模型架构图

2026 年 3 月 23 日

1 STAEformer 整体架构图

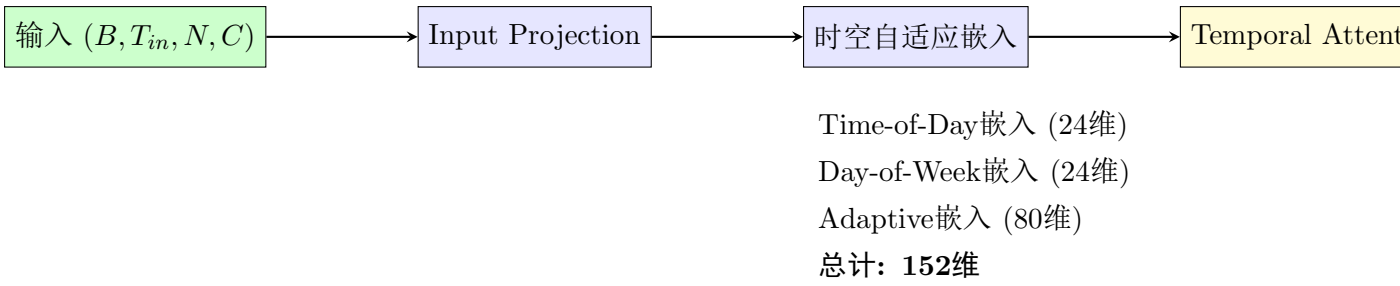
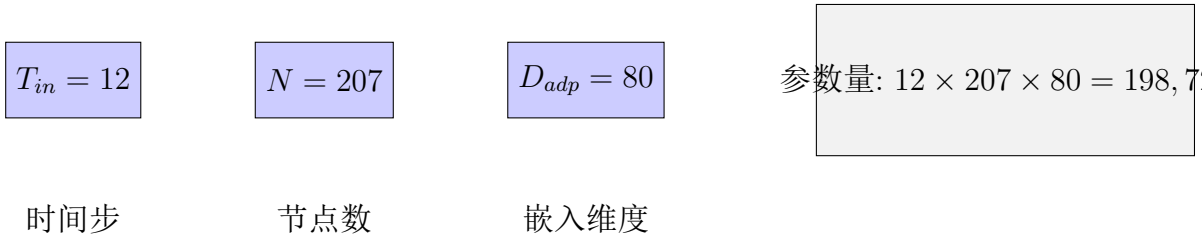
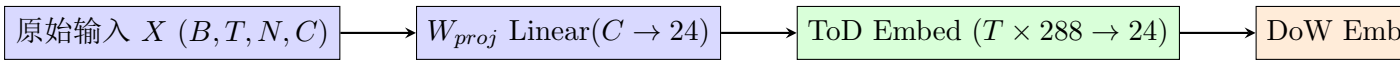


图 1: STAEformer 整体架构

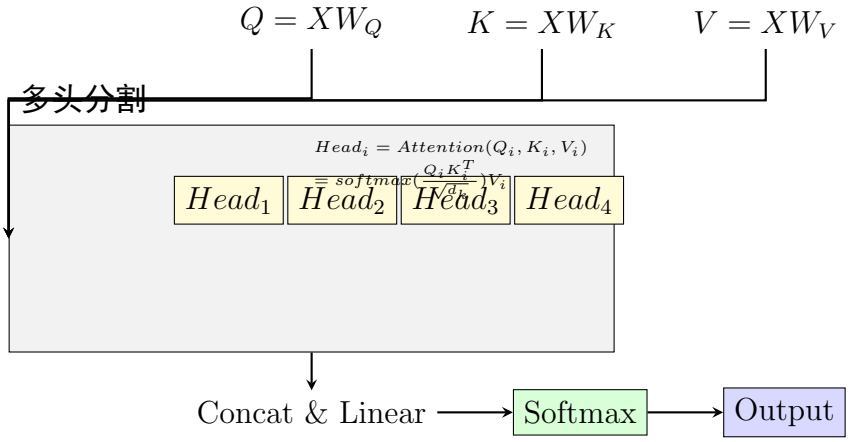
2 时空自适应嵌入机制


$T_{in} = 12$
$N = 207$
$D_{adp} = 80$
参数量: $12 \times 207 \times 80 = 198,720$
时间步
节点数
嵌入维度

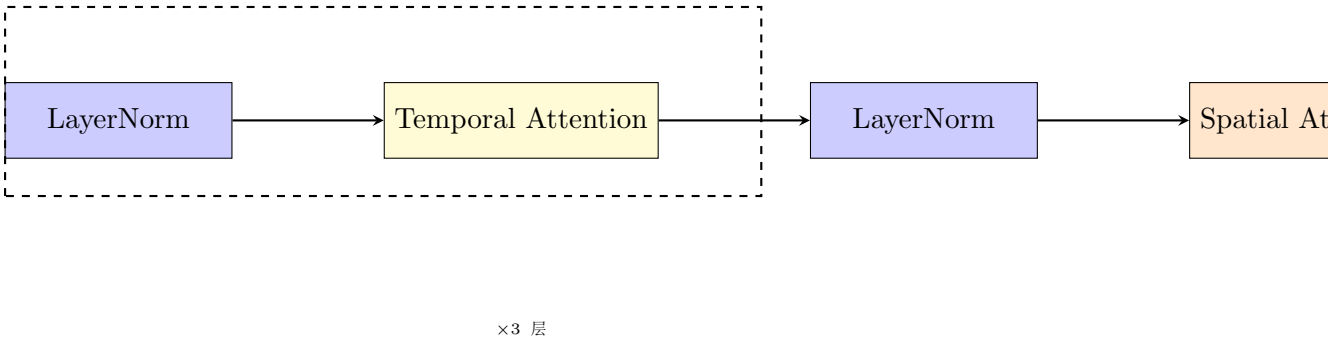
3 嵌入融合机制详解


原始输入 X (B, T, N, C)
W_{proj} Linear$(C \to 24)$
ToD Embed $(T \times 288 \to 24)$
DoW Emb

4 多头注意力机制


$Q = XW_Q$
$K = XW_K$
$V = XW_V$
多头分割
$Head_i = Attention(Q_i, K_i, V_i)$
$= softmax(\frac{Q_i K_i^T}{\sqrt{d_k}})V_i$
$Head_1$
$Head_2$
$Head_3$
$Head_4$
Concat & Linear
Softmax
Output

5 时空交替注意力



6 Mixed Projection 输出映射

